

Nathan (Kaiyan) Zhang

Vancouver, BC | nathanzhang1017@gmail.com | LinkedIn | GitHub | Website

Education

The University of British Columbia

B.A., Combined Major in Economics and Statistics; Minor in Data Science

Average: 93.5/100 cumulative

Selected coursework: Statistical Inference (100); Machine Learning and Data Mining (93); Causal Methods (98); Advanced Econometrics (98); Statistical Learning (97); Time Series and Forecasting (95)

Sep 2023 – May 2028

Vancouver, BC

Experience

Incoming ESG Co-op, Capital Markets

British Columbia Investment Management Corporation (BCI)

Sep 2026 – Dec 2026

Vancouver, BC

Research Assistant

UBC Sauder School of Business | Supervisor: Allen Hu

May 2026 – Present

Vancouver, BC

- Reconciled 1,330 RavenPack publication values against 3,210,045 ProQuest articles from Jul 1861 to May 2026, then audited coverage, missingness, provenance, and field reliability.
- Built the collection and validation pipeline for an 85,259-ID financial-creator directory, preserving raw media, metadata, run manifests, evidence links, and reconciliation results.

Research Assistant

UBC Sauder School of Business | Supervisors: J. Frank Li and Cheng Chen

Apr 2026 – Present

Vancouver, BC

- As the sole RA on a faculty-led study of age-biased technological change, linked CPS, PSID, NLSY, O*NET, and PATSTAT into worker, task, and patent panels. Built the wage and occupational sorting models.
- Harmonized 60+ O*NET releases and compared task text with about 56M PATSTAT records. Implemented share/log-share GMM, a 999-replication block bootstrap, and the R replication package.

Research Assistant

Vancouver School of Economics at UBC | Supervisors: Jonathan Graves and Laura Nelson

Jan 2025 – May 2026

Vancouver, BC

- Built Python/R learning modules and Quarto/GitHub Actions publishing for econometrics, forecasting, NLP, and computer vision, supporting 1,000+ cumulative student users.
- Designed, hosted, and taught a computer-vision workshop for social-science research. Coauthored an image-embedding notebook and a Programming Historian tutorial now under site-maintainer review.

Selected Research

Memory Is the Difference between Two Futures: Memory Response Paths and Path-Structured Context Compilation

Manuscript under review. With Jerrold Huang, designed a matched-future evaluation that removes one memory at a time and traces the difference over four horizons in 180 independent ecologies. In a separate 144-item evaluation, PSCC's exact-match accuracy exceeded A-Mem's by 6.94 percentage points (95% CI: 2.64 to 11.32) while using 67.2% of its API tokens.

When Answers Surprise

Manuscript under review. With Jerrold Huang, built and deduplicated a panel of about 31,000 firm events from 2018 to 2025, linking Capital IQ transcripts, IBES, CRSP, and historical S&P 500 membership to test whether management-answer novelty predicts returns beyond earnings surprise. We challenged the result with time holdouts, checks across learners, placebos, and timing tests.

Double Machine Learning for Payment-Friction Spending Effects

Course research project. Used cross-fitted DoubleML on 5,583 respondents and 22,332 in-person person-days to estimate the conditional effect of payment friction on daily spending. Under selection on observables and overlap, the estimated effect of lower-friction payment mixes was about 5%; heterogeneity was assessed by ticket size, income, and education.

Technical Skills

Programming and data: Python, R, SQL, Julia, DuckDB, Polars, Git/GitHub Actions, HPC/Slurm, Jupyter/Quarto

Modeling: causal inference/econometrics, panel/time series, machine learning, NLP, computer vision, scikit-learn, PyTorch, Transformers, fixest, statsmodels

Languages: Mandarin Chinese: native; English: academic working proficiency; Japanese: JLPT N3